Future Engineers

Sojourner

PRV Robotics

Yeriel Cruz
Aurelio Perez
Eliel Salas Morales

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Team Presentation

Describe in a paragraph your team members, roles, and responsibilities. Include a team photo.

- Names: Eliel, Aurelio, y Yeriel
- Ages: 17, 17 and 15
- Grades:11th, 11th and 10th
- Roles:
    - Eliel: YouTube Manager, Program and Building Supervisor, Note Taker, and Documenter.
    - Aurelio:field operations
    - Yeriel: Master Builder and Programmer.

Project Overview

Explain in two paragraphs what your robot does and its purpose in the competition.

- What does your robot do?

Our robot is an obstacle dodger meant to complete certain challenges within the established rules of the institution, the World Robot Olympiad, specifically in the Future Engineers category. It is designed to autonomously navigate a course while detecting and avoiding obstacles in real time. The system integrates sensors, programming logic, and mechanical design to ensure accuracy and efficiency during operation. By following competition guidelines, the robot demonstrates both innovation and adherence to engineering standards. Ultimately, this project reflects our ability to apply problem-solving skills and technological knowledge in a competitive robotics environment.

- Key features:

Our robot features use locomotive functions similar to those of a car, letting it move efficiently along the course. It uses two sensors to distinguish between colors and help pathfinding, enabling it to detect obstacles and follow the correct route. In terms of components, we incorporated two motors positioned at the front and back of the robot. These motors work together to propel the vehicle's wheels forward and backward with precision. This combination of sensors and motors allows the robot to navigate autonomously while maintaining control and stability.

Design Process & Initial Concepts

Describe in two paragraphs early ideas and designs you considered.

- Initial ideas:
- Why they were rejected:
- Final concept chosen:

We were going to go with a basic car design; however, it was rejected because it didn't work, so we ended up with an insanely complicated design with sensors for motion and sound. This robot is revolutionary (In this small school called E-Stream Academy)

System Overview

Provide a diagram and explain how all subsystems work together.

- Mechanical system
- Sensors
- Power
- Software

2 motors with 2 sensors

Lego spike software

7.2 voltage power supply

Estream Lego Sets

The robot is made out of 2 motors, 2 sensors, 4 wheels, and the power supply, powered by lego spike prime and our code blocks.

Design Tradeoffs

Explain in a paragraph important decision.

- We chose the EV3 brain rather than the Lego Spike because it proved more precise and efficient. We also chose turntables on the wheels instead of directly connecting them because this design increases stability. This setup reduces the risk of the wheels detaching during movement. As a result, the robot performs more reliably when navigating obstacles. Overall, these design choices improve both the efficiency and durability of the robot.

Components & Hardware

List and justify the components.

- Motors: it utilizes locomotive power from the LEGO spike sets in order to move and avoid obstacles and collisions.

- Sensors: sensors are utilized in order to measure movement, colors and track speed in order to keep consistency.

- Controller: there are no controllers, the robot is self controlled and will depend entirely upon the software in order to perform optimally.

- Other components: battery; this powers the entire system of the robot and is charged regularly to ensure that the robot runs smoothly and has high end performance.

Mobility & Mechanical Design

Explain chassis, wheels, and movement.

- Design description: chassis is made of bright Lego spike colors, highlighting its attractive and innovative design, while keeping the uninteresting parts out of sight.

- Diagrams/images:

- Testing and improvements: as we continued to run test to calibrate the accuracy of Sojourner, we used replica of the obstacle field, while repositioning the robot on the field various times to ensure that calibration would be effective. Thus creating the final product of our robot, Sojourner.

Power & Sensor Architecture

Explain power systems and sensors.

- Battery: so it has power
- Wiring diagram: horrendous
- Sensor placement: good
- Calibration: succesful

The Baterry is at the top, while the motors are at the bottom, and wiring diagram is all over the place and the calibration is just fine.

Software Architecture & Obstacle Strategy

Explain how the robot behaves.

- Flowchart/state machine:
- Lane following:
- Obstacle avoidance:
- Code explanation:

The robot behaves by pressing the play button on the proggram and making it move with the commands on the code in lego spike.

Testing & Iteration

Describe the testing process.

- Problems encountered: When we first began testing, we ran into problems. One of these problems was that the robot would often buffer before beginning to run, but this was easily solved by working with the coding in the spike program .

- Tests performed: the number of tests that were performed were around 8 times, each with their own unique calibration.

- Results: the calibration resulted in a higher rate of success when it came to the obstacle course.

- Improvements: we removed unnecessary features from the robot and enabled better performance.

Systems Thinking & Engineering Decisions

Explain overall system reasoning.

- Constraints: Our robot has difficulty precisely following the program as it can sometimes fall off course.
- Tradeoffs: We traded some of the potential speed for more accuracy, efficiency, and preciseness.
- Risk and solutions: The robot has a risk of losing power and precision if the battery starts to run out, our solution is to charge it before and after every exercise.

Reproducibility

Explain how someone can rebuild your robot.

- Build steps:
- Files included:
- GitHub structure:

People can rebuild our robot by looking for simple robot steps and looking through our github page and yt channel.

Future Improvements

Describe future upgrades.

For the future, we could upgrade the programming to make it more reliable and automated. This would allow the robot to respond to different challenges with little difficulty. We could also improve the wheel design to increase overall stability. Enhancing the wheels would make them more precise in movement. Additionally, stronger wheel attachments would reduce the chances of them falling off.

Sources

Lego Spike

WRO

List references used.
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